Smart Inventory Demand Forecasting Decision Support System: An Integrated Forecast-to-Decision Prototype

1st Kavyashri NP
Department of Computer Science and Engineering
CMR Institute of Technology
Bengaluru, India
kavyashri.np@cmrit.ac.in

2nd Pavan Raj B M
Department of Computer Science and Engineering
CMR Institute of Technology
Bengaluru, India
pabm23cs@cmrit.ac.in

3rd Pavan BV
Department of Computer Science and Engineering
CMR Institute of Technology
Bengaluru, India
pabv23cs@cmrit.ac.in

4th Ranjith Bhandari
Department of Computer Science and Engineering
CMR Institute of Technology
Bengaluru, India
rabh23cs@cmrit.ac.in

***Abstract*—Inventory planning requires a decision process that connects expected demand with current stock, supplier lead time, uncertainty, and managerial constraints. This paper presents a modular web-based Smart Inventory Demand Forecasting Decision Support System (DSS) that integrates supervised demand forecasting with inventory-risk calculations, scenario analysis, empirical uncertainty information, approximate feature-sensitivity analysis, and a human-in-the-loop approval workflow. The prototype is evaluated on a 48-observation chronological baseline dataset covering four products over twelve months. Product identity is one-hot encoded with lagged sales, inventory, unit price, month, and supplier lead time. Linear Regression, Ridge Regression, Decision Tree, and Random Forest are evaluated using a chronological 75:25 split, yielding 36 training and 12 holdout observations. Ridge Regression obtains the lowest holdout MAE of 13.39 and MAPE of 1.70%, while Linear Regression obtains the lowest RMSE of 16.58 and the highest R^2 of 0.9935. Ridge is therefore selected under the predefined MAE criterion. Because the dataset and holdout are small, these results are treated as preliminary evidence rather than proof of generalization. The system converts forecasts into safety-stock and reorder-point indicators, supports what-if analysis, and records approval decisions. The contribution is explicitly positioned as an integrated decision-support architecture rather than a new forecasting algorithm. A leakage-control protocol, rolling-origin validation plan, naive forecasting baseline, and operational inventory evaluation are identified as necessary extensions before production-level claims.**

***Index Terms*—Demand forecasting, inventory management, decision support system, machine learning, supply chain analytics, inventory optimization, human-in-the-loop, uncertainty.**

I. INTRODUCTION

Inventory decisions require a balance between product availability and the costs associated with holding, shortage, obsolescence, and delayed replenishment. A demand forecast is therefore useful only when it can be translated into an operational action under current inventory and supplier constraints. Recent research increasingly connects demand forecasting with inventory planning and control rather than treating forecasting as an isolated prediction problem [9]–[12].

Machine-learning methods have been applied to retail, e-commerce, and enterprise supply-chain forecasting. Poongothai *et al.* investigate machine learning for retail demand forecasting and inventory optimization [1], while Wang studies deep-learning demand prediction together with inventory optimization in e-commerce [2]. Zheng examines machine-learning algorithms for enterprise supply-chain demand forecasting [3], and Singh *et al.* study deep-learning approaches for forecasting and optimization in e-commerce supply chains [4]. Survey work further shows that forecasting performance depends strongly on data characteristics, forecasting horizon, and application context [7], [9], [10].

The forecast-to-inventory connection is especially important. van der Haar *et al.* investigate supervised learning for integrated forecasting and inventory control [12]. Sukolkit *et al.* connect demand forecasting to inventory management in the steel industry [11]. Liu *et al.* address inventory stockout prediction using machine learning [14], while An *et al.* study interpretable demand forecasting for micro-fulfillment environments [13]. Lingkon *et al.* investigate multi-item demand forecasting and downstream effects on replenishment, inventory cost, and service levels [15]. These studies motivate a system in which forecasting is only one stage of a broader decision process.

The practical gap addressed in this work is an integration gap. Existing research provides forecasting models, inventory policies, risk predictors, and interpretable analytical methods, but these components are frequently studied separately or in specialized operational settings. The proposed prototype combines forecasting, inventory indicators, scenario analysis,

approximate sensitivity information, and human authorization in one auditable workflow. It does not claim a new machine-learning algorithm.

The contributions are:

1) a layered forecast-to-decision architecture separating prediction from procurement authorization;
2) chronological comparison of four supervised regression models;
3) explicit integration of product identity and operational features;
4) empirical uncertainty and baseline-perturbation sensitivity information;
5) transparent safety-stock, reorder-point, and replenishment heuristics;
6) scenario-based what-if analysis; and
7) an auditable human-in-the-loop recommendation workflow.

II. RELATED WORK AND RESEARCH GAP

A. Demand Forecasting

Douaioui *et al.* critically review machine-learning and deep-learning approaches for demand forecasting in supply-chain management and identify substantial differences across datasets, horizons, and model families [9]. Goel *et al.* investigate demand forecasting using traditional machine learning and deep learning for different supply-chain contexts [10]. Zuvieta *et al.* survey statistical and machine-learning forecasting for aviation spare parts, emphasizing the difficulty of intermittent and non-smooth demand [7].

Application-specific studies provide additional evidence. Poongothai *et al.* investigate retail forecasting and inventory optimization [1]. Wang evaluates deep-learning demand prediction and inventory optimization in e-commerce [2]. Zheng studies machine-learning algorithms for enterprise supply-chain demand forecasting [3]. Singh *et al.* examine deep learning for forecasting and optimization in e-commerce supply chains [4]. Sathyabama *et al.* combine ARIMA and LSTM with expiry prediction and dynamic pricing for retail inventory management [5]. Manos *et al.* present an AI-based demand forecasting platform focused on supply-chain value [6].

B. Forecasting-to-Inventory Integration

Forecast accuracy alone does not determine an inventory action. van der Haar *et al.* explicitly connect supervised forecasting with inventory control [12]. Sukolkit *et al.* connect forecasting with inventory management in the steel industry [11]. Lingkon *et al.* investigate a hybrid multi-item forecasting approach and its implications for replenishment, inventory cost, and service levels [15]. These studies support the central design principle of the proposed DSS: the forecast should be an input to a transparent decision layer rather than the final output.

C. Risk and Interpretability

Liu *et al.* investigate machine learning for inventory stockout prediction, demonstrating the value of risk-oriented prediction beyond demand error alone [14]. An *et al.* investigate interpretable demand forecasting in micro-fulfillment environments [13]. These studies motivate the inclusion of risk indicators and approximate feature sensitivity.

D. Research Gap

The literature establishes substantial work on demand forecasting, inventory optimization, stockout prediction, and interpretability. The gap addressed here is not the absence of those methods. Instead, the prototype integrates them into an auditable workflow in which a forecast is transformed into inventory indicators, exposed to what-if analysis, and then presented to a human decision-maker. This integration is a systems contribution and should be evaluated primarily through reproducibility, decision transparency, and eventual operational outcomes rather than by claiming algorithmic novelty.

III. PROBLEM FORMULATION AND OBJECTIVES

Let D_t denote demand during period t, X_t the information available at the prediction origin, and $\hat{D}_{t+1} = f(X_t)$ the forecast for the next period. The decision-support layer maps the forecast and current operational state to an action:

$$\hat{D}_{t+1} = f(X_t), \qquad A_{t+1} = g(\hat{D}_{t+1}, I_t, L_t, U_t), \tag{1}$$

where I_t is inventory position, L_t is supplier lead time, U_t represents uncertainty information, and A_{t+1} is the recommended action. The current prototype keeps $g(\cdot)$ deterministic so that each recommendation can be traced to explicit assumptions.

The objectives are:

1) develop a modular web-based inventory DSS;
2) compare four supervised regression models using chronological evaluation;
3) preserve the temporal relationship between predictor information and target demand;
4) integrate product identity with operational variables;
5) expose empirical uncertainty and error information;
6) calculate safety stock and reorder point;
7) provide what-if scenario analysis; and
8) preserve human authority over procurement actions through an audit trail.

IV. SYSTEM ARCHITECTURE

The architecture separates presentation, authentication, application services, predictive analytics, decision logic, and persistence. This separation reduces coupling between the web application and model implementation and makes individual components replaceable.

TABLE I
RECENT RESEARCH AND RELATION TO THE PROPOSED DSS

Reference	Research focus	Evidence relevant to this work	Role in DSS
[9]	ML/DL supply-chain forecasting review	Model and application diversity	Literature foundation
[1]	Retail forecasting and inventory optimization	Forecasting connected to inventory	Retail context
[2]	E-commerce demand and inventory optimization	Deep-learning direction	Future model extension
[3]	Enterprise supply-chain forecasting	ML model application	Model comparison
[4]	E-commerce forecasting and optimization	Deep-learning supply-chain use	Future extension
[12]	Integrated forecasting and inventory control	Direct forecast-to-inventory link	Core motivation
[11]	Inventory-oriented demand forecasting	Forecast-to-inventory application	Inventory layer
[13]	Interpretable MFC forecasting	Interpretability in demand prediction	Sensitivity layer
[14]	Inventory stockout prediction	Risk-oriented prediction	Risk indicators
[15]	Multi-item supply-chain forecasting	Replenishment and service-level implications	Multi-SKU direction

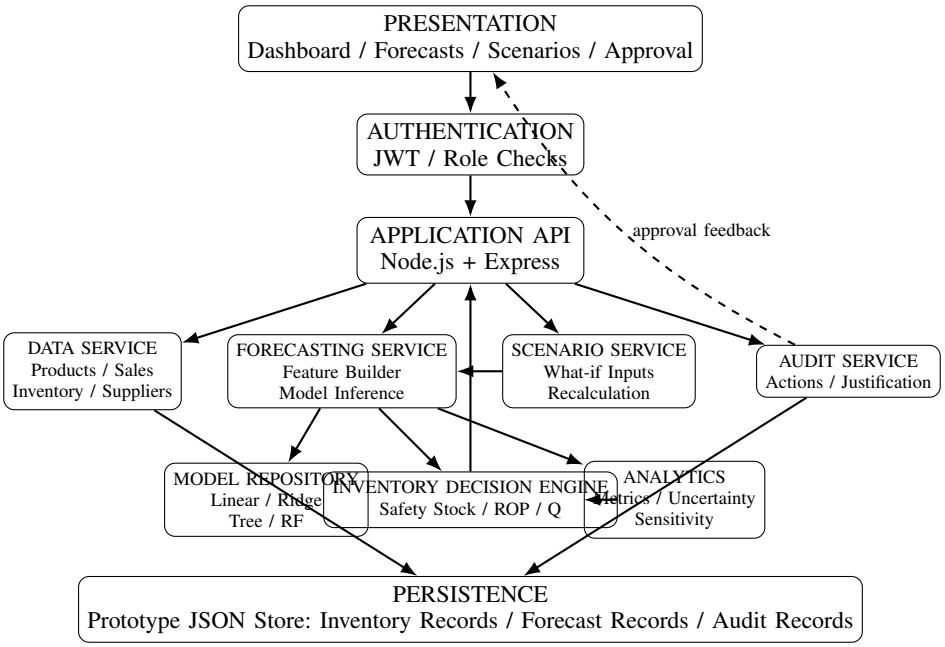


Fig. 1. Professional layered architecture of the Smart Inventory Demand Forecasting Decision Support System. Solid arrows represent primary application or data flow; dashed arrows represent audit feedback.

A. Presentation and Authentication

The presentation layer provides product selection, inventory status, demand forecasts, scenario analysis, and recommendation review. Authentication uses JSON Web Tokens (JWTs), while protected operations are separated from public application resources. The prototype does not claim enterprise-grade security.

B. Application and Forecasting Services

The Node.js and Express backend coordinates data retrieval, validation, feature preparation, model invocation, decision calculation, and audit recording. The forecasting service invokes serialized scikit-learn models. The separation permits the model implementation to evolve independently of the web interface.

C. Decision and Persistence Layers

The inventory decision engine converts forecasts into daily demand, safety stock, reorder point, and a prototype order quantity. Development persistence uses localized JSON records for products, inventory, suppliers, forecasts, and audit events. This choice is appropriate for a prototype but does not establish production-scale database performance.

V. METHODOLOGY

A. Dataset and Feature Construction

The baseline dataset contains 48 chronological observations covering four products over twelve months: Laptop Pro X1, Wireless Headphones, Smart Watch S3, and Gaming Mouse. Each observation contains product identity, previous-period sales, inventory position, unit price, month, supplier lead time, and target-period demand.

The supplied paper materials establish the dataset as a baseline prototype dataset but do not independently establish its external provenance. Therefore, this work makes no claim that the 48 observations are representative of commercial retail demand. The dataset should be distributed with the final research package if permitted.

Product identity is one-hot encoded. The feature vector is:

$$X_t = [P_t, S_{t-1}, I_t, Pr_t, M_t, L_t], \tag{2}$$

where S_{t-1} is lagged sales. The target is:

$$y_t = Demand_t. \tag{3}$$

The temporal interpretation requires all predictors to be available at the prediction origin. In particular, inventory must represent the stock position known before target-period demand is realized. If an inventory field was measured after target-period sales, it would constitute target leakage. This condition must be verified against the raw dataset before publication.

TABLE II
BASELINE DATASET AND FEATURE ROLES

Variable	Role
Product ID	One-hot encoded product identity
Previous sales	Lagged demand signal available before target period
Inventory	Inventory position at prediction origin
Unit price	Operational pricing signal
Month	Calendar-period indicator
Lead time	Supplier delivery-time information
Demand	Target variable

B. Data Quality and Preprocessing

The preprocessing stage checks schema consistency, missing values, duplicates, and numerical observations. Potential outliers are identified using:

$$IQR = Q_3 - Q_1. \tag{4}$$

Observations below $Q_1 - 1.5IQR$ or above $Q_3 + 1.5IQR$ are flagged. Twelve price observations were flagged in the supplied baseline data and retained because they were treated as legitimate observations rather than automatically deleted.

Continuous variables used by Linear Regression and Ridge Regression are standardized using training-set statistics:

$$x' = \frac{x - \mu_{train}}{\sigma_{train}}. \tag{5}$$

The same parameters are applied to holdout observations. Tree-based models do not require scaling.

C. Chronological Holdout

The first nine monthly observations for each product are used for training and the final three for testing:

$$N_{train} = 4 \times 9 = 36, \qquad N_{test} = 4 \times 3 = 12. \tag{6}$$

No random shuffling is used. This split reflects the intended deployment direction. However, a single 12-observation holdout is not sufficient to establish robust temporal generalization. Consequently, the reported results are treated as preliminary.

D. Forecasting Models

Four supervised regression models are evaluated. Linear Regression provides a transparent baseline. Ridge adds L_2 regularization. Decision Tree models nonlinear threshold relationships, while Random Forest aggregates multiple trees.

Linear Regression minimizes:

$$\min_w ||Xw - y||_2^2. \tag{7}$$

Ridge Regression minimizes:

$$\min_w ||Xw - y||_2^2 + \alpha||w||_2^2, \tag{8}$$

with $\alpha = 1.0$.

The Decision Tree uses maximum depth five. Random Forest uses 100 trees.

TABLE III
MODEL CONFIGURATIONS

Model	Configuration
Linear Regression	fit intercept
Ridge Regression	$\alpha = 1.0$
Decision Tree	maximum depth = 5
Random Forest	100 trees

E. Evaluation Metrics

The models are evaluated using:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{9}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{10}$$

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|, \tag{11}$$

and:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{12}$$

MAE is the primary model-selection criterion because it expresses error directly in demand units. RMSE exposes larger errors, while MAPE and R^2 provide complementary views. If the target contains zero-demand periods, MAPE must be replaced or protected with an appropriate zero-safe formulation.

F. Uncertainty and Sensitivity

For Random Forest, the standard deviation of individual tree predictions is used as an empirical dispersion estimate:

$$\sigma_{trees} = Std(f_1(x), \ldots, f_K(x)). \tag{13}$$

An empirical normal-approximation band is:

$$U_{90} = \hat{y} \pm 1.645\sigma_{trees}. \tag{14}$$

This is explicitly an empirical uncertainty band and is not claimed to be a calibrated 90% prediction interval.

For Ridge, uncertainty is estimated from chronological training residuals. Feature sensitivity uses baseline perturbation: one feature is changed while remaining features are held at a baseline vector, and the resulting prediction difference is recorded. This is a model-specific sensitivity measure and is not claimed to provide formal SHAP attribution.

G. Inventory Decision Formulation

Forecasted monthly demand is converted to approximate daily demand:

$$\bar{d} = \frac{\hat{D}}{30}. \tag{15}$$

The generalized safety-stock expression is:

$$SS = Z\sqrt{\bar{L}\sigma_d^2 + \bar{d}^2\sigma_L^2}. \tag{16}$$

When lead-time variability is negligible:

$$SS = Z\sigma_d\sqrt{\bar{L}}. \tag{17}$$

The prototype uses $Z = 1.65$. The reorder point is:

$$ROP = \bar{d}\bar{L} + SS. \tag{18}$$

The prototype replenishment quantity is:

$$Q = \max(0, \hat{D} + SS - I), \tag{19}$$

where I is current inventory. This is a transparent replenishment heuristic, not a cost-optimal inventory-control policy.

VI. Robustness and Reproducibility Protocol

A publication-level forecasting evaluation should not rely on a single small holdout. The current experiment therefore distinguishes between the *reported baseline experiment* and the *required robustness protocol*.

First, a naive persistence baseline should be evaluated:

$$\hat{D}_t = D_{t-1}. \tag{20}$$

This determines whether the ML models outperform a simple last-period-demand rule.

Second, rolling-origin validation should be performed. For forecast origins $t_1, \ldots, t_m$, the model is repeatedly trained only on observations available before each origin and evaluated on the subsequent observation. Mean and standard deviation of MAE, RMSE, and MAPE should then be reported. This avoids relying on a single arbitrary test window.

Third, leakage must be audited for every feature. In particular:

$$S_{t-1}, I_t, Pr_t, M_t, L_t \rightarrow D_t \tag{21}$$

is valid only when each variable is available before D_t is realized. Any post-demand inventory or sales variable must be excluded.

Fourth, per-product results should be reported. Aggregate metrics can hide poor performance for an individual SKU. A stronger final experiment should therefore report MAE, RMSE, and MAPE for each of the four products.

These robustness procedures are not represented as completed experiments unless the underlying dataset and code reproduce them. They are included as explicit controls against overinterpretation of the current 12-observation holdout.

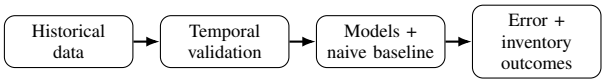


Fig. 2. Recommended reproducibility and robustness evaluation sequence.

VII. Experimental Results

The baseline experiment uses the 48-row dataset and the 12-observation chronological holdout. The reported environment is Node.js 18.x, Python 3.10, pandas 2.0, and scikit-learn 1.2.

TABLE IV
MODEL PERFORMANCE ON THE 12-OBSERVATION HOLDOUT

Model	MAE	RMSE	MAPE	R^2
Random Forest	48.10	60.30	5.97%	0.9140
Linear Regression	13.87	**16.58**	2.00%	**0.9935**
Ridge Regression	**13.39**	17.70	**1.70%**	0.9926
Decision Tree	44.72	54.34	5.58%	0.9304

Ridge obtains the lowest MAE and MAPE. Linear Regression obtains the lowest RMSE and highest R^2. Ridge is therefore selected according to the predefined MAE criterion; it is not claimed to dominate every metric.

The difference between Ridge and Linear Regression in MAE is small, while RMSE favors Linear Regression. This illustrates why model selection should be connected to the operational loss function. If average absolute error is the principal concern, MAE provides an interpretable selection rule. If large errors carry disproportionate cost, RMSE or an asymmetric inventory-cost function would be more appropriate.

The tree-based models perform substantially worse on this small holdout. This does not establish that tree methods are generally inferior. The result may reflect the limited number of observations and the absence of richer nonlinear temporal signals. Hyperparameter tuning on a larger dataset could change the ranking.

A. Interpretation of High R^2

The linear models produce R^2 values above 0.99. Such values are unusually high for a small forecasting experiment. Possible explanations include strong structure in the baseline

dataset, highly informative lagged features, or leakage in feature construction. The current evidence is therefore insufficient to generalize these values to commercial demand forecasting.

The final research package should preserve the raw dataset, preprocessing code, prediction outputs, model configurations, and evaluation scripts. Independent reproduction should be performed before making strong accuracy claims.

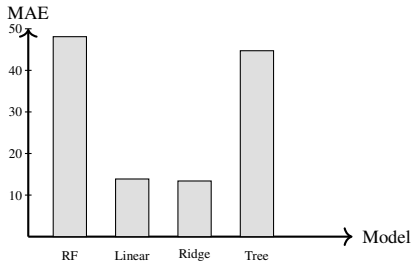


Fig. 3. Reported holdout MAE values. The figure is descriptive of the 12-observation baseline experiment and should not be interpreted as evidence of broad generalization.

VIII. DECISION SUPPORT AND IMPLEMENTATION

A. Scenario Analysis

The scenario module permits users to modify selected assumptions such as price, lead time, inventory, and demand-related inputs before accepting a forecast. Modified inputs are evaluated without permanently changing historical training records. Scenario outputs are therefore decision-support evidence rather than training-data modifications.

B. Human-in-the-Loop Recommendation Workflow

The recommendation workflow separates prediction from authorization. A forecast can generate a recommendation, but the recommendation enters a pending state before any procurement action is authorized. The operator can choose APPROVE, REJECT, or SNOOZE. Rejected or delayed recommendations can include a contextual reason. The audit record stores the timestamp, user identifier, selected model, forecast, recommendation, action, and justification.

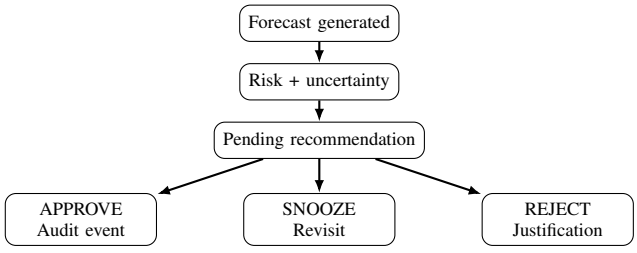


Fig. 4. Human-in-the-loop recommendation and audit workflow.

C. Security and Audit

JWT authentication restricts protected operations to authenticated users. Audit records support accountability by retaining recommendation outcomes and user justifications. The prototype does not claim enterprise-grade security. Production deployment would require secure secret management, rate limiting, centralized logging, input validation, stronger authorization policies, and security testing.

D. Reproducibility

A reproducible release should include the dataset, feature definitions, preprocessing rules, chronological split, model configurations, prediction outputs, evaluation scripts, software versions, random seeds for stochastic models, model hashes, and dataset versions. The current paper documents the primary experimental configuration but does not claim that all of these artifacts are included in the supplied submission package.

IX. OPERATIONAL EVALUATION FRAMEWORK

Forecast metrics measure prediction error, but the stated objective of the DSS is to improve inventory decisions. Therefore, a complete evaluation should include an operational comparison between the proposed forecast-driven policy and a baseline replenishment policy.

The following indicators are recommended:

$$ServiceLevel = 1 - \frac{StockoutDemand}{TotalDemand}, \tag{22}$$

together with stockout frequency, average inventory, inventory turnover, replenishment frequency, and total inventory cost.

A cost-oriented evaluation can be represented by:

$$C_{total} = C_h I_{avg} + C_o N_o + C_s D_{stockout}, \tag{23}$$

where C_h is holding cost per inventory unit, I_{avg} is average inventory, C_o is ordering cost, N_o is the number of orders, C_s is shortage cost, and $D_{stockout}$ is stockout demand.

These quantities are not reported as completed experimental results in the current study because the supplied baseline does not establish the cost parameters required for a defensible cost comparison. Including the framework makes the distinction between predictive performance and operational performance explicit.

X. LIMITATIONS AND THREATS TO VALIDITY

The principal limitation is dataset size. The baseline contains only 48 observations and four products, with twelve observations in the holdout. The experiment therefore cannot establish robust long-term seasonality, holiday effects, structural changes, or commercial-scale generalization.

A second limitation is validation design. One chronological holdout is appropriate as a basic temporal test but is weaker than rolling-origin evaluation. A third limitation is dataset provenance: the supplied paper materials do not independently establish that the baseline observations represent a real commercial inventory environment. Claims should therefore remain at prototype level.

A fourth threat is target leakage. The temporal use of lagged sales is explicitly designed to avoid direct target leakage, but the timing semantics of inventory and other operational variables must be verified against the raw data. A fifth limitation is the absence of a naive forecasting baseline in

the reported experiment. Without such a baseline, it cannot be concluded that the machine-learning models outperform a simple persistence rule.

The uncertainty estimate is not calibrated, and baseline perturbation does not provide formal SHAP attribution. The inventory layer uses a deterministic replenishment heuristic rather than a cost-optimal policy. JSON persistence is suitable for development but does not demonstrate production scalability. Finally, the unusually high linear-model R^2 values require stronger temporal validation before deployment.

XI. FUTURE RESEARCH

Future work should first strengthen the empirical foundation rather than simply add more complex models. A larger multi-year dataset with more SKUs, promotions, holidays, stockout indicators, and external demand drivers should be evaluated using rolling-origin validation.

The model comparison should include a naive persistence baseline and, where seasonality is established, a seasonal-naive baseline. Statistical uncertainty across forecast origins should be reported. Per-product performance should be examined to identify SKUs for which aggregate metrics hide systematic errors.

The inventory layer should be evaluated against a conventional replenishment policy using service level, stockout frequency, average inventory, inventory turnover, and total cost. Holding, ordering, shortage, and service-level parameters should be explicitly defined before optimization claims are made.

Future versions can investigate gradient boosting, LSTM, GRU, or transformer models when sufficient data are available. Probabilistic forecasting and conformal prediction can replace heuristic uncertainty bands. Formal explainability methods can be evaluated after increasing dataset size. The persistence layer can be migrated to a relational database and tested under realistic concurrent workloads.

A production-oriented version should additionally support scheduled retraining, model versioning, data-drift monitoring, automated data-quality checks, role-based approval policies, and reproducible model artifacts.

XII. CONCLUSION

This paper presented a Smart Inventory Demand Forecasting Decision Support System that integrates supervised demand forecasting, inventory decision logic, scenario analysis, empirical uncertainty, approximate feature sensitivity, and human approval. On the supplied 48-observation baseline dataset, Ridge Regression obtained the lowest holdout MAE of 13.39 and MAPE of 1.70%, while Linear Regression obtained the lowest RMSE of 16.58 and highest R^2 of 0.9935.

The results are preliminary because the holdout contains only twelve observations and because the supplied materials do not independently establish broad external representativeness of the baseline dataset. The principal contribution is therefore the integration of forecasting and inventory decision support into an auditable workflow, not a claim of a new forecasting algorithm.

For publication-quality evidence, the next evaluation should verify feature timing and leakage, add a naive baseline, use rolling-origin validation, report per-product performance, and evaluate operational inventory outcomes. Subject to these validations, the proposed architecture provides a structured foundation for translating demand predictions into transparent and human-controlled inventory decisions.

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